The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Canonical Square-Root Operator Core
from a Hidden-Response Hessian
Minimization Law

Abstract

The current active-primary theorem already proves that the current carrier-bearing canonical normalized selected-response lift collapses benchmark-facingly to a smaller canonical square-root operator core $\mathfrak{O}^{\mathrm{disp,body,sqrt}}$ on the fixed displacement fibers. The next honest benchmark-facing move is therefore no longer another replay of the square-root operator core itself, of the normalized-lift package, of the selected-lift package, of the body-level forcing law, or of the visible-response structures that follow from them. It is to derive or justify the current square-root operator core from still more primitive explicit substrate structure.

The present note makes that move in the cleanest forced-readout form presently available below the square-root-operator-core frontier. For each sector, it introduces a benchmark-faithful hidden-response Hessian minimization law: the fixed proto-section benchmark base and fixed section-centered displacement body together with a finite-dimensional hidden-response carrier, a smooth surjective readout to the displacement fibers, and a smooth positive self-adjoint hidden Hessian on that carrier. The current square-root operator core is then the canonical reduced readout of that deeper package. Writing

$$M_\sigma^{\mathrm{disp,hid}}(y) \equiv \Pi_\sigma^{\mathrm{disp,hid}}(y) A_\sigma^{\mathrm{disp,hid}}(y)^{-1} \Pi_\sigma^{\mathrm{disp,hid}}(y)^*, \qquad B_\sigma^{\mathrm{disp,hid}}(y) \equiv M_\sigma^{\mathrm{disp,hid}}(y)^{-1},$$

the induced constrained minimization energy is exactly

$$\frac{1}{2}\left\langle v, B_\sigma^{\mathrm{disp,hid}}(y) v\right\rangle = \frac{1}{2}\left\langle R_\sigma^{\mathrm{disp,body,sqrt}}(y) v, R_\sigma^{\mathrm{disp,body,sqrt}}(y) v\right\rangle, \qquad R_\sigma^{\mathrm{disp,body,sqrt}}(y) \equiv B_\sigma^{\mathrm{disp,hid}}(y)^{1/2}.$$

Hence the current canonical square-root operator core is no longer primitive once this deeper hidden-response Hessian minimization law is fixed.

The benchmark boundary remains unchanged throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. By the current square-root-operator-core frontier, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, the same current body-response core, and every currently recorded downstream benchmark-facing consequence then follow unchanged. The remaining honest burden therefore moves below the current square-root operator core to a deeper derivation or justification of the hidden-response Hessian minimization law that forces it.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has already removed the larger stopping points that once sat above the current section-centered hidden-response datum: the visible-response operator-family presentation, the body-level forcing law, the operator-bearing selected-lift package, and the carrier-bearing normalized selected-response lift. The current active-primary theorem shows that the remaining benchmark-relevant content of those presentations is already carried by a smaller canonical square-root operator core on the fixed displacement fibers.

The next honest question is therefore narrower again. What still more primitive explicit substrate structure forces that current square-root operator core itself? The present note answers that question in a deliberately disciplined way. It does not restore the larger normalized-lift package, the operator-bearing selected-lift package, the current-body image core, a larger reduced-carrier presentation, the older projection/body presentation, or a replay of the full displacement-response substrate law. Instead it retains only the data needed to pose a constrained hidden-response minimization problem over the actual displacement fibers: a hidden carrier, a displacement readout, and a positive hidden Hessian. The current square-root operator core is then the canonical reduced readout of that deeper package.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim a completed first-principles derivation of GR from final substrate microphysics. Its narrower positive role is to show that the current canonical square-root operator core is already a canonical output of a deeper hidden-response Hessian minimization law.

2. Fixed Inputs from the Current Square-Root-Operator-Core Frontier

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Definition 2 (Recorded benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active route already fixes the following data and consequences.*

1. *the current proto-section benchmark base, in particular the recorded proto-section body* $\mathcal{B}^{\mathrm{psec}}_{\sigma}$ *and shell* $\mathcal{Z}^{\mathrm{psec}}_{\sigma}$*;*

2. *the fixed section-centered displacement body*

$$\mathcal{D}^{\mathrm{pppresec}}_{\sigma} \subset \mathcal{B}^{\mathrm{psec}}_{\sigma} \times V^{\mathrm{pppresec}}_{\sigma}$$

whose fiber

$$\mathcal{D}^{\mathrm{pppresec}}_{\sigma}(y) \equiv \{v \in V^{\mathrm{pppresec}}_{\sigma} : (y, v) \in \mathcal{D}^{\mathrm{pppresec}}_{\sigma}\}$$

is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}^{\mathrm{psec}}_{\sigma}$*;*

3. *the recorded downstream continuation, namely*

$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,res}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec}},$$
$$\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec,sec}},$$

together with the current continuity-Hessian compressed core and benchmark one-metric observer package;

4. *benchmark faithfulness, meaning preservation of the benchmark outputs* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Proposition 1 (What the current square-root-operator-core frontier already proves)**.** *At current scope, the already-recorded square-root-operator-core frontier theorems prove all of the following.*

1. *Any benchmark-faithful canonical square-root operator core canonically determines the current section-centered displacement-body quadratic forcing law on the fixed displacement body by*

$$\mathscr{F}^{\mathrm{disp,resp,vis,body}}_{\sigma}(y, v) = \frac{1}{2}\left\langle R^{\mathrm{disp,body,sqrt}}_{\sigma}(y)v, R^{\mathrm{disp,body,sqrt}}_{\sigma}(y)v \right\rangle.$$

2. *That law canonically determines the visible-response operator family* $S^{\mathrm{disp,resp,vis}}_{\sigma}$ *and the canonical minimal visible-response representative* $\mathfrak{S}^{\mathrm{disp,resp,red,min}}_{\sigma}$.

3. *The current body-response core and every currently recorded downstream benchmark-facing consequence then follow unchanged from the same induced forcing law.*

Proof. Item (1) is exactly the content of the current theorem collapsing the canonical normalized

selected-response lift to the canonical square-root operator core. Items (2) and (3) are exactly the forcing-law downstream continuation already preserved on the recorded route. □

Definition 3 (Benchmark-faithful canonical square-root operator core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful canonical square-root operator core *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*
2. *A smooth family of positive self-adjoint square-root operators*
$$R_\sigma^{\mathrm{disp,body,sqrt}}(y) : V_\sigma^{\mathrm{pppresec}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$
3. *Positivity on the actual displacement body:*
$$\left\langle R_\sigma^{\mathrm{disp,body,sqrt}}(y)v, R_\sigma^{\mathrm{disp,body,sqrt}}(y)v \right\rangle = 0 \qquad \Longleftrightarrow \qquad v = 0$$
for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ *and every* $v \in \mathcal{D}_\sigma^{\mathrm{pppresec}}(y)$*.*
4. *Benchmark faithfulness in the sense of Definition 2(4).*

3. Hidden-Response Hessian Minimization Law

Definition 4 (Benchmark-faithful hidden-response Hessian minimization law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*. A* benchmark-faithful hidden-response Hessian minimization law *for sector* σ *consists of the following data.*

1. *The fixed proto-section benchmark base and fixed section-centered displacement body of Definition 2(1)–(2).*
2. *A finite-dimensional Euclidean hidden-response carrier*
$$\mathcal{H}_\sigma^{\mathrm{disp,hid}}.$$
3. *A smooth family of surjective linear hidden-response readouts*
$$\Pi_\sigma^{\mathrm{disp,hid}}(y) : \mathcal{H}_\sigma^{\mathrm{disp,hid}} \to V_\sigma^{\mathrm{pppresec}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}}.$$
4. *A smooth family of positive self-adjoint hidden Hessians*
$$A_\sigma^{\mathrm{disp,hid}}(y) : \mathcal{H}_\sigma^{\mathrm{disp,hid}} \to \mathcal{H}_\sigma^{\mathrm{disp,hid}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$
each of which is positive definite on $\mathcal{H}_\sigma^{\mathrm{disp,hid}}$*.*
5. *Benchmark faithfulness in the sense of Definition 2(4).*

Remark 1. *Definition 4 is intentionally smaller than the older current-body-image-core, reduced-projection-response-structure, and projection/body presentations. It keeps only the hidden carrier, the displacement readout, and the positive hidden Hessian needed to formulate the constrained minimization problem. It does not reinstall the normalized-lift package, the operator-bearing selected-lift*

package, the body-level forcing law, the visible-response operator family, the canonical minimal visible-response representative, the current body-response core, or the larger image-core and projection/body packages as independently primitive data.

4. Canonical Reduced Readout

Proposition 2 (Positive reduced Schur operator)**.** *Fix a benchmark-faithful hidden-response Hessian minimization law for sector σ. For each $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, define*

$$M_\sigma^{\mathrm{disp,hid}}(y) \equiv \Pi_\sigma^{\mathrm{disp,hid}}(y) A_\sigma^{\mathrm{disp,hid}}(y)^{-1} \Pi_\sigma^{\mathrm{disp,hid}}(y)^* : V_\sigma^{\mathrm{pppresec}} \to V_\sigma^{\mathrm{pppresec}}, \tag{2}$$

where $(\cdot)^*$ denotes Euclidean adjoint with respect to the fixed inner products on $\mathcal{H}_\sigma^{\mathrm{disp,hid}}$ and $V_\sigma^{\mathrm{pppresec}}$. Then $M_\sigma^{\mathrm{disp,hid}}(y)$ is positive self-adjoint and invertible.

Proof. Self-adjointness is immediate from the self-adjointness of $A_\sigma^{\mathrm{disp,hid}}(y)^{-1}$. For any $0 \neq w \in V_\sigma^{\mathrm{pppresec}}$,

$$\left\langle w, M_\sigma^{\mathrm{disp,hid}}(y) w \right\rangle = \left\langle A_\sigma^{\mathrm{disp,hid}}(y)^{-1/2} \Pi_\sigma^{\mathrm{disp,hid}}(y)^* w, A_\sigma^{\mathrm{disp,hid}}(y)^{-1/2} \Pi_\sigma^{\mathrm{disp,hid}}(y)^* w \right\rangle > 0$$

because $\Pi_\sigma^{\mathrm{disp,hid}}(y)$ is surjective and therefore $\Pi_\sigma^{\mathrm{disp,hid}}(y)^*$ is injective. Hence $M_\sigma^{\mathrm{disp,hid}}(y)$ is positive definite and therefore invertible on the finite-dimensional space $V_\sigma^{\mathrm{pppresec}}$. □

Definition 5 (Canonical minimizer lift and reduced operator)**.** *Fix a benchmark-faithful hidden-response Hessian minimization law for sector σ. For each $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, define the* canonical hidden-response minimizer lift

$$L_\sigma^{\mathrm{disp,hid,min}}(y) \equiv A_\sigma^{\mathrm{disp,hid}}(y)^{-1} \Pi_\sigma^{\mathrm{disp,hid}}(y)^* M_\sigma^{\mathrm{disp,hid}}(y)^{-1} : V_\sigma^{\mathrm{pppresec}} \to \mathcal{H}_\sigma^{\mathrm{disp,hid}} \tag{3}$$

and the reduced hidden-response operator

$$B_\sigma^{\mathrm{disp,hid}}(y) \equiv M_\sigma^{\mathrm{disp,hid}}(y)^{-1} : V_\sigma^{\mathrm{pppresec}} \to V_\sigma^{\mathrm{pppresec}}. \tag{4}$$

Proposition 3 (Canonical minimizer lift properties)**.** *For every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$,*

1. $$\Pi_\sigma^{\mathrm{disp,hid}}(y) \circ L_\sigma^{\mathrm{disp,hid,min}}(y) = \mathrm{id}_{V_\sigma^{\mathrm{pppresec}}};$$

2. *if $K_\sigma^{\mathrm{disp,hid}}(y) \equiv \ker \Pi_\sigma^{\mathrm{disp,hid}}(y)$, then*

   $$\left\langle L_\sigma^{\mathrm{disp,hid,min}}(y) v, A_\sigma^{\mathrm{disp,hid}}(y) k \right\rangle = 0$$

   for every $v \in V_\sigma^{\mathrm{pppresec}}$ and every $k \in K_\sigma^{\mathrm{disp,hid}}(y)$;

3. *every $u \in \mathcal{H}_\sigma^{\mathrm{disp,hid}}$ admits a unique decomposition*

   $$u = L_\sigma^{\mathrm{disp,hid,min}}(y)\big(\Pi_\sigma^{\mathrm{disp,hid}}(y) u\big) + k, \qquad k \in K_\sigma^{\mathrm{disp,hid}}(y);$$

4. *$L_\sigma^{\mathrm{disp,hid,min}}(y)$ is linear in the displacement argument and smooth in y.*

Proof. Item (1) is

$$\Pi_\sigma^{\mathrm{disp,hid}}(y)L_\sigma^{\mathrm{disp,hid,min}}(y) = \Pi_\sigma^{\mathrm{disp,hid}}(y)A_\sigma^{\mathrm{disp,hid}}(y)^{-1}\Pi_\sigma^{\mathrm{disp,hid}}(y)^* M_\sigma^{\mathrm{disp,hid}}(y)^{-1} = M_\sigma^{\mathrm{disp,hid}}(y)M_\sigma^{\mathrm{disp,hid}}(y)^{-1} = \mathrm{id}.$$

For item (2), let $k \in K_\sigma^{\mathrm{disp,hid}}(y)$. Then

$$\left\langle L_\sigma^{\mathrm{disp,hid,min}}(y)v, A_\sigma^{\mathrm{disp,hid}}(y)k \right\rangle = \left\langle M_\sigma^{\mathrm{disp,hid}}(y)^{-1}v, \Pi_\sigma^{\mathrm{disp,hid}}(y)k \right\rangle = 0.$$

Item (3) follows by setting

$$k \equiv u - L_\sigma^{\mathrm{disp,hid,min}}(y)(\Pi_\sigma^{\mathrm{disp,hid}}(y)u)$$

and applying item (1). Uniqueness follows because item (1) fixes the non-kernel part. Linearity and smoothness are immediate from (3). □

Proposition 4 (Unique hidden-response energy minimizer)**.** *For every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ and every $v \in V_\sigma^{\mathrm{ppresec}}$, the canonical hidden-response minimizer lift $L_\sigma^{\mathrm{disp,hid,min}}(y)v$ is the unique minimizer of*

$$\min\left\{\frac{1}{2}\left\langle u, A_\sigma^{\mathrm{disp,hid}}(y)u\right\rangle : u \in \mathcal{H}_\sigma^{\mathrm{disp,hid}},\ \Pi_\sigma^{\mathrm{disp,hid}}(y)u = v\right\}. \tag{5}$$

Proof. Let $u \in \mathcal{H}_\sigma^{\mathrm{disp,hid}}$ satisfy $\Pi_\sigma^{\mathrm{disp,hid}}(y)u = v$. By Proposition 3(3),

$$u = L_\sigma^{\mathrm{disp,hid,min}}(y)v + k, \qquad k \in K_\sigma^{\mathrm{disp,hid}}(y).$$

Using Proposition 3(2),

$$\left\langle u, A_\sigma^{\mathrm{disp,hid}}(y)u\right\rangle = \left\langle L_\sigma^{\mathrm{disp,hid,min}}(y)v, A_\sigma^{\mathrm{disp,hid}}(y)L_\sigma^{\mathrm{disp,hid,min}}(y)v\right\rangle + \left\langle k, A_\sigma^{\mathrm{disp,hid}}(y)k\right\rangle.$$

The last term is strictly positive for every nonzero k because $A_\sigma^{\mathrm{disp,hid}}(y)$ is positive definite. Therefore the unique minimizer of (5) is $L_\sigma^{\mathrm{disp,hid,min}}(y)v$. □

Definition 6 (Hidden-response minimization energy)**.** *Fix $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ and a benchmark-faithful hidden-response Hessian minimization law. Define the induced hidden-response minimization energy on the recorded displacement body by*

$$\mathcal{E}_\sigma^{\mathrm{hid,min}}(y,v) \equiv \min\left\{\frac{1}{2}\left\langle u, A_\sigma^{\mathrm{disp,hid}}(y)u\right\rangle : u \in \mathcal{H}_\sigma^{\mathrm{disp,hid}},\ \Pi_\sigma^{\mathrm{disp,hid}}(y)u = v\right\} \tag{6}$$

for every $(y,v) \in \mathcal{D}_\sigma^{\mathrm{ppresec}}$.

Proposition 5 (Closed form of the hidden-response minimization energy)**.** *For every $(y,v) \in \mathcal{D}_\sigma^{\mathrm{ppresec}}$,*

$$\mathcal{E}_\sigma^{\mathrm{hid,min}}(y,v) = \frac{1}{2}\left\langle L_\sigma^{\mathrm{disp,hid,min}}(y)v, A_\sigma^{\mathrm{disp,hid}}(y)L_\sigma^{\mathrm{disp,hid,min}}(y)v\right\rangle = \frac{1}{2}\left\langle v, B_\sigma^{\mathrm{disp,hid}}(y)v\right\rangle. \tag{7}$$

Proof. The first equality is Proposition 4. For the second,

$$\left\langle L_\sigma^{\mathrm{disp,hid,min}}(y)v, A_\sigma^{\mathrm{disp,hid}}(y)L_\sigma^{\mathrm{disp,hid,min}}(y)v\right\rangle = \left\langle A_\sigma^{\mathrm{disp,hid}}(y)^{-1}\Pi_\sigma^{\mathrm{disp,hid}}(y)^* B_\sigma^{\mathrm{disp,hid}}(y)v, \Pi_\sigma^{\mathrm{disp,hid}}(y)^* B_\sigma^{\mathrm{disp,hid}}(y)v\right\rangle$$

and hence

$$\left\langle L_\sigma^{\mathrm{disp,hid,min}}(y)v, A_\sigma^{\mathrm{disp,hid}}(y)L_\sigma^{\mathrm{disp,hid,min}}(y)v\right\rangle = \left\langle \Pi_\sigma^{\mathrm{disp,hid}}(y)A_\sigma^{\mathrm{disp,hid}}(y)^{-1}\Pi_\sigma^{\mathrm{disp,hid}}(y)^* B_\sigma^{\mathrm{disp,hid}}(y)v, B_\sigma^{\mathrm{disp,hid}}(y)v\right\rangle$$

□

Proposition 6 (Canonical square-root readout)**.** *Fix a benchmark-faithful hidden-response Hessian minimization law for sector σ. Define*

$$R_\sigma^{\mathrm{disp,body,sqrt}}(y) \equiv B_\sigma^{\mathrm{disp,hid}}(y)^{1/2} \tag{8}$$

for each $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$. Then $R_\sigma^{\mathrm{disp,body,sqrt}}$ is a benchmark-faithful canonical square-root operator core and

$$\mathcal{E}_\sigma^{\mathrm{hid,min}}(y,v) = \frac{1}{2}\left\langle R_\sigma^{\mathrm{disp,body,sqrt}}(y)v, R_\sigma^{\mathrm{disp,body,sqrt}}(y)v\right\rangle \tag{9}$$

for every $(y,v) \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}$.

Proof. By Proposition 2, $B_\sigma^{\mathrm{disp,hid}}(y) = M_\sigma^{\mathrm{disp,hid}}(y)^{-1}$ is positive self-adjoint, so it has a unique positive self-adjoint square root $R_\sigma^{\mathrm{disp,body,sqrt}}(y)$. Smooth dependence on y follows from the smooth dependence of $B_\sigma^{\mathrm{disp,hid}}(y)$ and the smooth functional calculus in finite dimension.

For $(y,v) \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}$,

$$\left\langle R_\sigma^{\mathrm{disp,body,sqrt}}(y)v, R_\sigma^{\mathrm{disp,body,sqrt}}(y)v\right\rangle = \left\langle v, B_\sigma^{\mathrm{disp,hid}}(y)v\right\rangle$$

by definition of the positive square root. Combined with Proposition 5, this proves (9). Because $B_\sigma^{\mathrm{disp,hid}}(y)$ is positive definite, that quadratic form vanishes only at $v = 0$, so positivity on the actual displacement body holds. Benchmark faithfulness is inherited from Definition 4(5). Therefore the data satisfy the definition of a benchmark-faithful canonical square-root operator core. □

5. Deriving the Current Square-Root-Operator-Core Frontier

Theorem 1 (Current canonical square-root operator core from a hidden-response Hessian minimization law)**.** *Fix the primitive continuation data of Definition 1, the recorded benchmark base and downstream chain of Definition 2, and for each sector $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ a benchmark-faithful hidden-response Hessian minimization law in the sense of Definition 4. Then the current canonical square-root operator core is canonically forced by that deeper explicit substrate package. More precisely:*

1. *Propositions 2, 5, and 6 canonically associate to the hidden-response Hessian minimization law a reduced operator $B_\sigma^{\mathrm{disp,hid}}$ and a benchmark-faithful square-root operator core $R_\sigma^{\mathrm{disp,body,sqrt}}$.*

2. *The induced hidden-response minimization energy is exactly the current square-root-operator-core energy:*
$$\mathcal{E}_\sigma^{\mathrm{hid,min}}(y,v) = \frac{1}{2}\left\langle R_\sigma^{\mathrm{disp,body,sqrt}}(y)v, R_\sigma^{\mathrm{disp,body,sqrt}}(y)v\right\rangle.$$

3. *By Proposition 1, the current section-centered displacement-body quadratic forcing law, the same visible-response operator family $S_\sigma^{\mathrm{disp,resp,vis}}$, the same canonical minimal visible-response representative $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$, the same current body-response core, and every currently*

recorded downstream benchmark-facing consequence then follow unchanged from that same hidden-response Hessian minimization law through the induced square-root operator core.

4. *The benchmark one-metric package* (1) *is preserved, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

5. *Consequently the remaining honest burden is no longer to restate the current square-root operator core itself, but to derive or justify the hidden-response Hessian minimization law that forces it from still more primitive explicit substrate structure.*

Proof. Item (1) is exactly Propositions 2, 5, and 6. Item (2) is equation (9). Item (3) is the previously recorded square-root-operator-core downstream continuation applied to the forcing law furnished by Item (2). Item (4) is the benchmark-faithfulness condition built into Definitions 2 and 4. Item (5) is the routing consequence of Items (1)–(4): once the current square-root operator core is shown to be a canonical output of the deeper hidden-response Hessian minimization law, another manuscript that merely restates that same square-root operator core or merely replays the same downstream visible-response consequences does not remove any remaining benchmark-facing freedom. □

Corollary 1 (The square-root operator core is no longer primitive)**.** *Under the hypotheses of Theorem 1, the current canonical square-root operator core is no longer a primitive stopping point on the recorded section-centered hidden-response route. Its benchmark-relevant content is already forced by the deeper hidden-response Hessian minimization law* $\mathfrak{M}^{\mathrm{disp,hid,min}}$.

Proof. Theorem 1(1)–(3) identifies the current square-root-operator-core energy, the current body-level forcing law, and the same downstream benchmark-facing consequences as outputs of the hidden-response Hessian minimization law. Therefore the current square-root operator core is no longer primitive at benchmark scope once that deeper law is fixed. □

6. Conclusion

The positive result of this note is a genuine derivational gain below the current square-root-operator-core frontier. The current canonical square-root operator core no longer needs to be treated as primitive. It is already a canonical reduced readout of a deeper hidden-response Hessian minimization law on the fixed displacement fibers.

The scope remains narrow and honest. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and nothing here upgrades adoption to a completed first-principles derivation of GR from final substrate microphysics.

The next open burden is therefore sharper again. The next benchmark-facing manuscript should derive or justify the hidden-response Hessian minimization law itself from still more primitive explicit substrate structure. Another manuscript that merely restates the same hidden-response minimization package, the same square-root operator core, the same carrier-bearing normalized selected-response lift, the same operator-bearing selected-lift package, the same section-centered displacement-body quadratic forcing law, the same visible-response operator family, the same canonical minimal visible-response representative, the same current body-response core, or a larger same-output current-image-core or reduced-carrier or projection/body or full substrate-law presentation is no longer the clean next target at current scope.